

Alae Herrak

Full-Stack Software Engineer

✉ alaeherrak@gmail.com | 📍 Fes, Morocco | 🌐 Portfolio | 🐙 alae-herrak | 🌐 alae-herrak

Summary

I am a full-stack software engineer with over 3 years of experience building web platforms and desktop applications. My primary work involves TypeScript, React, Next.js, and Node.js, along with desktop runtimes like Tauri and Electron. I focus on building reliable internal tools, clear data models, and maintainable software for business and administrative workflows.

Work Experience

Whois	Fes, Morocco (Hybrid)
Full-Stack Software Engineer	Aug 2024 – Sep 2026
- Built and maintained web applications and desktop tools for public administration, education, and retail clients. - Packaged cross-platform desktop applications using Tauri v2 and Electron, setting up automated GitHub Actions release builds and updates. - Handled financial data modeling with BigInt arithmetic and built Excel import workflows (SheetJS) to prevent calculation drift in public budget lines. - Designed relational database schemas and built REST and WebSocket APIs using Bun, Node.js, Express, and Zod. - Created client-side document export workflows supporting both French and Arabic (RTL) layouts.	
Smart Services Team	Fes, Morocco (Freelance)
Full-Stack Software Engineer	Oct 2023 – Aug 2024
- Developed web applications and admin dashboards using React, TypeScript, and Tailwind CSS with bilingual RTL support. - Designed MySQL schemas and REST APIs with Node.js and Express for grant intake and inventory tracking. - Built desktop POS tools using Electron with local receipt printer and barcode scanner integration. - Created multi-step forms with input validation to ensure consistent data intake.	

Featured Production Systems

Yosan - Public Procurement & Budget Execution ERP
- Built a desktop ERP to track multi-tier public procurement budgets and expenditure lines. - Implemented fixed-precision calculations using BigInt and added Excel import workflows to load structured budget sheets.
alaeherrak.com/projects#yosan-budget
SmartHire - AI Screening & Recruitment Platform
- Developed a recruitment platform combining local LLM data extraction with deterministic scoring rules. - Added file hashing to detect duplicate candidate submissions and streamline applicant reviews.
alaeherrak.com/projects#smarthire
Project Qarawiyyin - Higher Education Management Ecosystem
- Developed an internal management desktop app for university administration and student records serving 3000+ daily active users. - Implemented role-based access control and live state updates using WebSockets.
alaeherrak.com/projects#qarawiyyin
Storyland - Multi-Tenant Gamified Reading Platform
- Built a multi-tenant reading platform with an interactive canvas map to track student reading milestones. - Implemented a timed quiz engine with automated grading and attempt limits.
alaeherrak.com/projects#storyland-edtech

Education

École Nationale des Sciences Appliquées de Fès	Master's degree
Data Science Engineering and Artificial Intelligence	
Specialized Institute Of Applied Technology	Specialized Technician
Full Stack Development	

Technical Skills

Languages & Frontend
TypeScript, JavaScript, PHP, Python, HTML5/CSS3, React, Next.js, Tailwind CSS, Vite
Backend & Runtimes
Node.js, Bun, Express, Laravel, FastAPI, REST APIs, WebSockets, Socket.IO, Zod
Systems & Tools
Tauri v2, Electron, PostgreSQL, MySQL, MongoDB, Prisma ORM, Git, GitHub Actions, Docker